INCH-POUND

MIL-G-80275
8 March 1988

MILITARY SPECIFICATION

GRINDING MACHINES, CIRCULAR SAW BLADE,
METAL-CUTTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, horizontal and vertical spindle metal-cutting circular saw blade grinding machines with the equipment and accessories specified herein.

1.2 Classification. The grinding machines shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 3.5 and 6.2.1).

Type I - Horizontal spindle.

Size	Saw blade diameter range (inches)
1	8 to 30
2	24 to 60
3	24 to 72
4	24 to 100

Type II - Vertical spindle.

Size	Saw blade diameter range (inches)
1	8 to 30

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3415

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2. APPLICABLE REFERENCE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. The issue of these documents shall be the latest revision of the document on the date of the solicitation and may include title changes.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. The issue shall be the issue in effect on the date of the solicitation and may include title changes.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification as specified. The issues of these documents shall be the latest revision of the documents on the date of the solicitation and may include title changes.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1	- Motors and Generators.
ANSI/NFPA 79	- Electrical Standard for Industrial Machinery.
ANSI B7.1	- Safety Requirements for the Use, Care, and Protection of Abrasive Wheels.
ANSI B11.9	- Safety Requirements for the Construction, Care, and Use of Grinding Machines.
ANSI B74.2	- Specifications for Shapes and Sizes of Grinding Wheels and Shapes, Sizes, and Identification of Mounted Wheels.
ANSI B74.13	- Markings for Identifying Grinding Wheels and Other Bonded Abrasives.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	- Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02	- Manual for Machine Tool Gearing.
AGMA 390.03	- Gear Classification, Materials, and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Modules and Diametrical Pitches of Cylindrical Gears for General Engineering and for Heavy Engineering.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of reproducing metal-cutting circular saw tooth shapes. Means shall be provided to manually set up the machine and control saw blade and grinding wheel contact. The machine shall maintain uniform hook and tooth angles or restore saw teeth points on saw blades within the saw blade diameter capacity of the machine. Type I machines shall have a horizontally mounted wheel spindle; type II machines shall have a vertically mounted wheel spindle. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified (see 6.2.1), either the U. S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in either US or SI units as specified by procuring activity (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

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3.2.2 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.01 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position. When specified (see 6.2.1), indicating dials shall be graduated to read measurements in both US and SI units. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for the point of operation and all other parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of ANSI B11.9 and 29 CFR 1910 that are applicable to the machine itself. When the requirements of the ANSI B11.9 and 29 CFR 1910 standards conflict, the 29 CFR 1910 standards shall take precedence. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

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3.2.10 Interchangeability. Replacement parts (spare or repair parts) shall be manufactured to definite dimensions and tolerances. Tooling and accessories normally supplied as standard equipment shall also be replaceable. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. All parts subject to future replacement as spare or repair parts shall be assigned a part number identified in the repair parts manual. Foreign sourced replacement parts shall be interchangeable with domestically supplied components or be readily available from a domestic source.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts, either US or SI, used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears, shafts, and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the US system or ANSI/ISO 54 for the SI system. The gears, shafts, and pinions shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

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3.3.8 Slides. All sliding members shall be provided with means to meet and maintain the accuracy and alignment tolerances given in table II.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of a base, grinding wheelhead assembly, saw blade holder, drive components, and controls. Type I machines shall be provided with a tooth shape generating mechanism and type II machines shall be provided with a universal workhead to index saw blades and grind various tooth shapes. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Base. The base shall be a boxlike construction of cast iron or steel alloy, ribbed and braced to withstand forces imposed when the machine is used to its maximum rated capacity. The base shall support the grinding wheelhead assembly, the saw blade holder, the drive mechanisms and controls, and other related machine components. The base on type I machines shall house the tooth shape generating mechanism and driving components. An access door(s) or removable plate(s) shall be provided for servicing components housed within the base. Means for floor mounting and leveling the machine shall be provided. When necessary, a sub-base shall be provided to allow grinding large diameter saw blades.

3.4.2 Grinding wheelhead assembly. The grinding wheelhead assembly shall be mounted to the base, composed of the wheelhead, spindle, drive mechanism, and related components. The grinding wheelhead assembly shall oscillate to grind tooth faces and gullets. On type I machines, the wheelhead shall be coordinated with the tooth shape generating mechanism (see 3.4.4) so the action of that mechanism will cause the proper saw blade shape to be ground. On type II machines, the wheelhead shall be coordinated with the universal workhead to grind various tooth geometries.

3.4.2.1 Grinding wheelhead. The wheelhead shall be properly guarded and house the grinding wheel spindle and related components. The wheelhead spindle shall rotate in the wheelhead assembly on bearings and accept a driving mechanism at one end and a grinding wheel at the other end. The wheelhead driving mechanism shall be powered by an electric motor. Unless otherwise specified (see 6.2.1), the drive shall be through a belt and pulley(s). Means for adjusting belt tension shall be provided unless a nontensioning type belt is used. Grinding wheel requirements shall be in accordance with table I.

3.4.2.2 Grinding wheels. Unless otherwise specified (see 6.2.1), grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All grinding wheels shall be provided with guards.

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3.4.3 Saw blade holder. Type I machines shall be provided with a saw blade holder mounted to the base to accept saw blades within the diameter range for the size machine specified. Means for mounting and holding the saw blade on the holder shall be provided. The holder shall be adjustable to accommodate saw blades of various diameters. The saw blade mounting device shall permit 360 degree indexing of the saw blade. An adjustable clamp or support shall be provided to prevent saw blade vibration during grinding.

3.4.4 Universal workhead. Type II machines shall be provided with a universal workhead for holding the saw blade. The workhead shall be mounted on a vertically adjustable table on the base. The universal workhead shall accept saw blades within the diameter range specified in table I and shall be adjustable to allow grinding various tooth shapes. Means for mounting and holding the blade shall be provided. The workhead shall allow indexing the saw blade 360 degrees. Means shall be provided to prevent saw blade vibration during grinding.

3.4.5 Tooth shape generating mechanism. The tooth shape generating mechanism on type I machines shall be housed in the base and be easily accessible. The mechanism shall reproduce required tooth shapes through the grinding wheelhead assembly to the saw blade. The arrangement shall include means for interchangeability to provide for a series of tooth shapes. The tooth shape generating mechanism shall be adjustable for various tooth spacings and depths. Dials or scales shall indicate tooth hook angles. Unless otherwise specified (see 6.2.1), a variable lift feature shall be incorporated in the feed system. Means shall be provided, through the variable lift, to adjust the effect of the lift cam to allow either deep or shallow teeth to be sharpened without changing cams. Unless otherwise specified (see 6.2.1), the tooth shape generating mechanism shall be powered by an electric motor, independent from the grinding wheelhead assembly, and shall be driven through a belt and pulley(s). Means for adjusting belt tension shall be provided unless a nontensioning belt is used. Unless otherwise specified (see 6.2.1), the tooth shape generating mechanism and its feed system shall meet the requirements given in table I.

3.4.6 Hydraulic system. The hydraulic system, when provided, shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.7 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit having a fused safety disconnect switch or circuit breaker. Initial input voltage shall be as specified (see 6.2.1). An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

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3.4.7.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), alternating-current (AC) motors shall be designed to operated on 60-Hz and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1.

3.4.7.2 Dust extractor. When specified (see 6.2.1), a dust extractor to be used during dry grinding shall be provided. The dust extractor shall be connected to the machine's electrical system with start/stop pushbuttons conveniently located on the control panel.

3.4.7.3 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than specified in table I for the machine size specified.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II when tested according to the procedures given in 4.8.1.

3.7 Performance. The machine shall be capable of grinding saw blades in accordance with the requirements herein and specifically in accordance with the test requirements designated in 4.8.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of special wrenches, operating, and repair tools normally furnished by the manufacturer.

b. One adjustable worklight, complete with protective shield, on-off switch, and mounting hardware.

c. One set of grinding wheels normally furnished by the manufacturer (not less than 3).

d. Tapered cones or bushings in increments not to exceed 1/8 inch for saw blades with arbor holes for the range given in table I for the size machine specified.

e. Tooth shape generating cams for type I machines; the number of each, type, style, and all other essential information as designated by the procuring activity (see 6.2.1).

f. One extra set of wheelhead drive belts.

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g. One extra set of tooth shape generating mechanism drive belts for type I machines.

h. Wheel dresser, permanently mounted or independent (see 6.2.1).

3.9 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

a. Additional grinding wheels: shank size, number of each wheel size, type dimensional characteristics, and composition of each wheel, as specified (see 6.2.1).

b. Other optional equipment as specified.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1). All repair parts shall be assigned a part number identified in a repair parts manual.

3.11 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity
Military or Federal specification number for each lubricant
Size and type of each lubricant filter

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

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3.12 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.7).

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with an ambient temperature of 50°F to 100°F with 10 percent to 95 percent relative humidity.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspections shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the first article examination and all tests to be accepted.

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4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspections shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. The acceptance test shall be performed on the machine to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification including compliance with all referenced documents in Section 2.

4.8 Tests.

4.8.1 Alignment and accuracy test. Each machine and components shall be checked for compliance with the alignment and accuracy requirements specified in 3.6. Alignment and accuracy tests shall be performed before and after performing the tests given in 4.8.2 through 4.8.3.1. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine.

4.8.2 Operational test. The machine shall be operated at no load for not less than 1 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Performance test. The machine shall be subjected to the following performance test. All instruments, materials, and tools required to perform and evaluate the tests shall be furnished by the supplier. The measuring instruments used shall have evidence of calibration traceable to the National Bureau of Standards (NBS). At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following test (see 6.2.1).

4.8.3.1 Saw blade grinding test. Each machine shall be subjected to a grinding test. Saw blade(s) shall be ground in accordance with the requirements specified by the procuring activity (see 6.2.1) including the number of blades, range in gage, range in diameters, tooth spacing from point to point, tooth hook angles, tooth style(s), diameter of saw eyes, acceptability criteria, and other necessary requirements for testing machine performance. Unless otherwise specified (see 6.2.1), the procuring activity shall furnish the saw blade(s) to be ground. The method used to perform the test shall be at the supplier's option provided it accurately determines the capability of the machine.

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4.8.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packaging, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in any shop where metal cutting circular saw blades are used and need to be resharpened.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1.1).

f. Dual dials, if required (see 3.2.2).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

j. Grinding wheelhead drive, if different (see 3.4.2.1).

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k. Grinding wheels, if different (see 3.4.2.2).

l. Variable lift, if not required (see 3.4.5).

m. Tooth shape generating mechanism motor and drive requirements, if different (see 3.4.5).

n. Tooth shape generating mechanism characteristics, if different (see 3.4.5).

o. Electrical system, if different; input voltage, as specified (see 3.4.7).

p. Motor, if different (see 3.4.7.1).

q. Dust extractor, if required (see 3.4.7.2).

r. Hour meter, if required (see 3.4.7.3).

s. Size and capacity, if different (see 3.5).

t. Standard equipment, if different (see 3.8).

u. Number, type, style, and other information for tooth shape generating cams for type I machines (see 3.8.(e)).

v. Wheel dresser, as specified (see 3.8.(h)).

w. Specify and fully describe optional equipment, if required (see 3.9).

x. Additional grinding wheels, type, dimensional characteristics, and composition required (see 3.9.(a)).

y. Specify applicable repair parts, if required (see 3.10).

z. Lubrication plate, if different (see 3.11.1).

aa. Nameplate, if different (see 3.11.2).

bb. Inspection conditions, if different (see 4.3).

cc. First article inspection, if different (see 4.4).

dd. Quality conformance inspection, if different (see 4.5).

ee. Acceptance test, if different (see 4.6).

ff. Performance test, if different (see 4.8.3).

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gg. Geometry of saw blade(s) to be ground for performance test (diameter, tooth spacing, tooth depth, tooth hook angles, tooth style, and saw eye diameter)(see 4.8.3.1).

hh. Saw blade(s) to be ground to be furnished by manufacturer, if required (see 4.8.3.1).

ii. If preservation, packing, and marking in accordance with MIL-M-18058 are required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts, lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (keyword) listing.

Circular saw blade, metalworking
Grinder, circular saw blade
Machine tool, grinding, circular saw
Metalworking, saw blade
Saw blade, circular
Wheel, abrasive, grinding

6.5 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.6 Warranty. Warranty requirements shall be as specified in the contract by the procuring activity.

6.7 Inspection facilities. The contracting officer should be notified by each contractor in his response to the solicitation of the location where inspection tests are to be performed. It is recommended the contract requires all inspections be performed within the limits of the continental United States.

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Table I. Sizes and capacities.

Characteristic	Requirement			
	Types I and II	Type I		
	Size 1	Size 2	Size 3	Size 4
Capacity:				
Sawblade diameter range, inches	8 to 30	24 to 60	24 to 72	24 to 100
Feed, teeth per minute	25 to 29			
Tooth spacing, maximum, inches	1	2	2	3-1/2
Tooth depth, gullet, inches	1	2	2	3
Tooth hook, positive degrees	0 to 60			
Saw hole range, inches	1-1/4 to 6	3-1/2 to 18	3-1/2 to 18	3-1/2 to 24
Grinding wheel, inches				
Diameter	8			12
Bore	3/4			
Thickness	*	*	*	*
Speed, surface feet per minute (SFPM)	6500			7200
Motors, HP:				
Wheelhead drive	3/4	1		
Tooth generating mechanism	1/2	3/4		

NOTE 1: Asterisks (*) indicate wheel thickness shall be provided as required by the procuring activity.

NOTE 2: Machine sizes and capacities shall not be less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in the above table. When the largest size and capacity machine shown in the above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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Table II. Alignment and accuracy requirements.

Test to be applied	Permissible tolerance
Tooth positioning accuracy	± 0.003 inch
Grinding wheel spindle runout	± 0.001 T.I.R.
Grinding wheelhead repeatability	± 0.002 inch

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - CR, ER, SM
Navy - SH
Air Force - 84
DLA - GS

User activities:
Army - AR, ME, SM
Navy - MC, AS

Preparing activity:
DLA - IP

Project (3415-0167)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80275	Grinding Machines, Circular Saw Blade, Metalcutting

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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